



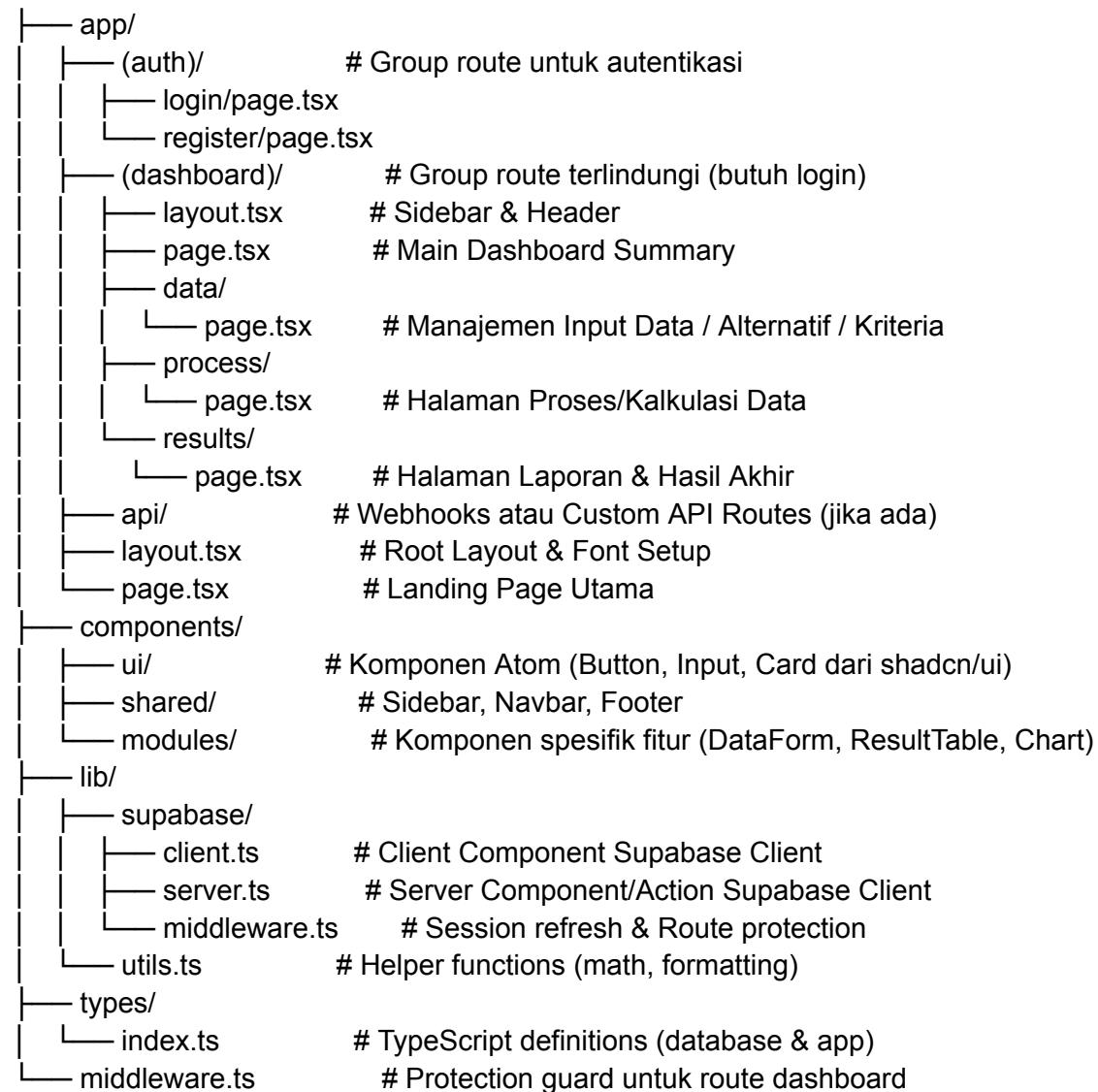
SPESIFIKASI SISTEM & BLUEPRINT ARSITEKTUR

Target Stack:

- **Frontend & Backend:** Next.js (App Router, Server Actions/Route Handlers, Tailwind CSS, shadcn/ui)
- **Database & Auth:** Supabase (PostgreSQL, Row Level Security / RLS, Auth SSR)
- **Hosting/Deployment:** Vercel

1. Arsitektur Folder & Struktur Aplikasi (Next.js App Router)

Plaintext



2. Skema Database Supabase (PostgreSQL + RLS)

Jalankan skrip SQL ini langsung di **Supabase SQL Editor** untuk membuat tabel, relasi, dan keamanan data berbasis pengguna (*Row Level Security*).

SQL

-- 1. Tabel Profil Pengguna (Tersinkronisasi dengan Supabase Auth)

```
CREATE TABLE public.profiles (  
  id UUID REFERENCES auth.users ON DELETE CASCADE PRIMARY KEY,  
  full_name TEXT,  
  role TEXT DEFAULT 'user' CHECK (role IN ('admin', 'user')),  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT TIMEZONE('utc'::text, NOW()) NOT  
  NULL  
);
```

-- Turn on RLS for profiles

```
ALTER TABLE public.profiles ENABLE ROW LEVEL SECURITY;
```

CREATE POLICY "Pengguna dapat melihat profil sendiri"

```
ON public.profiles FOR SELECT  
USING (auth.uid() = id);
```

CREATE POLICY "Pengguna dapat memperbarui profil sendiri"

```
ON public.profiles FOR UPDATE  
USING (auth.uid() = id);
```

-- 2. Tabel Proyek / Kasus Keputusan (Projects / Datasets)

```
CREATE TABLE public.projects (  
  id UUID DEFAULT gen_random_uuid() PRIMARY KEY,  
  user_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE NOT NULL,  
  title VARCHAR(255) NOT NULL,  
  description TEXT,  
  status VARCHAR(50) DEFAULT 'draft' CHECK (status IN ('draft', 'processed', 'archived')),  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT TIMEZONE('utc'::text, NOW()) NOT  
  NULL,  
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT TIMEZONE('utc'::text, NOW()) NOT  
  NULL  
);
```

```
ALTER TABLE public.projects ENABLE ROW LEVEL SECURITY;
```

CREATE POLICY "User hanya bisa mengelola proyek miliknya"

```
ON public.projects FOR ALL  
USING (auth.uid() = user_id);
```

-- 3. Tabel Kriteria Evaluation (Criteria)

```
CREATE TABLE public.criteria (  
  id UUID DEFAULT gen_random_uuid() PRIMARY KEY,  
  project_id UUID REFERENCES public.projects(id) ON DELETE CASCADE NOT NULL,  
  code VARCHAR(50) NOT NULL,          -- Contoh: C1, C2
```

```

name VARCHAR(255) NOT NULL,          -- Contoh: Biaya, Kualitas
weight NUMERIC(5,4) NOT NULL,        -- Bobot kriteria (misal: 0.25)
type VARCHAR(10) CHECK (type IN ('benefit', 'cost')) NOT NULL,
created_at TIMESTAMP WITH TIME ZONE DEFAULT TIMEZONE('utc'::text, NOW()) NOT
NULL
);

```

```

ALTER TABLE public.criteria ENABLE ROW LEVEL SECURITY;

```

```

CREATE POLICY "User mengelola kriteria via proyeknya"
ON public.criteria FOR ALL
USING (
    EXISTS (
        SELECT 1 FROM public.projects
        WHERE projects.id = criteria.project_id AND projects.user_id = auth.uid()
    )
);

```

-- 4. Tabel Alternatif Data (Alternatives)

```

CREATE TABLE public.alternatives (
    id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
    project_id UUID REFERENCES public.projects(id) ON DELETE CASCADE NOT NULL,
    code VARCHAR(50) NOT NULL,          -- Contoh: A1, A2
    name VARCHAR(255) NOT NULL,         -- Contoh: Opsi A, Kandidaat B
    created_at TIMESTAMP WITH TIME ZONE DEFAULT TIMEZONE('utc'::text, NOW()) NOT
NULL
);

```

```

ALTER TABLE public.alternatives ENABLE ROW LEVEL SECURITY;

```

```

CREATE POLICY "User mengelola alternatif via proyeknya"
ON public.alternatives FOR ALL
USING (
    EXISTS (
        SELECT 1 FROM public.projects
        WHERE projects.id = alternatives.project_id AND projects.user_id = auth.uid()
    )
);

```

-- 5. Tabel Nilai Matriks Evaluasi (Evaluation Values)

```

CREATE TABLE public.evaluations (
    id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
    alternative_id UUID REFERENCES public.alternatives(id) ON DELETE CASCADE NOT
NULL,
    criterion_id UUID REFERENCES public.criteria(id) ON DELETE CASCADE NOT NULL,
    value NUMERIC(10,4) NOT NULL,
    UNIQUE(alternative_id, criterion_id)
);

```

```
ALTER TABLE public.evaluations ENABLE ROW LEVEL SECURITY;
```

```
CREATE POLICY "User mengelola matriks evaluasi miliknya"
ON public.evaluations FOR ALL
USING (
  EXISTS (
    SELECT 1 FROM public.alternatives
    JOIN public.projects ON projects.id = alternatives.project_id
    WHERE alternatives.id = evaluations.alternative_id AND projects.user_id = auth.uid()
  )
);
```

-- 6. Tabel Hasil Perhitungan (Calculation Results)

```
CREATE TABLE public.results (
  id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
  project_id UUID REFERENCES public.projects(id) ON DELETE CASCADE NOT NULL,
  alternative_id UUID REFERENCES public.alternatives(id) ON DELETE CASCADE NOT
NULL,
  score NUMERIC(10,6) NOT NULL,
  rank INT NOT NULL,
  calculated_at TIMESTAMP WITH TIME ZONE DEFAULT TIMEZONE('utc'::text, NOW())
NOT NULL
);
```

```
ALTER TABLE public.results ENABLE ROW LEVEL SECURITY;
```

```
CREATE POLICY "User melihat hasil kalkulasi miliknya"
ON public.results FOR ALL
USING (
  EXISTS (
    SELECT 1 FROM public.projects
    WHERE projects.id = results.project_id AND projects.user_id = auth.uid()
  )
);
```

3. Konsep UI / UX & Desain Halaman

Halaman	Deskripsi & Komponen Utama	Akses Kontrol
Landing Page (/)	Hero Section, penjelasan fitur utama, CTA Login/Register.	Publik

Auth (/login, /register)	Form login/daftar dengan email + password atau OAuth (Google).	Publik
Dashboard (/page.tsx)	Metrics ringkasan (Jumlah Proyek, Status Analisis), list proyek terbaru.	Authenticated
Data Input (/data)	Tabbed Interface: 1. Input Kriteria & Bobot, 2. Input Alternatif, 3. Matriks Nilai (Tabel Interaktif).	Authenticated
Proses Analisis (/process)	Tombol "Jalankan Kalkulasi", animasi loading indikator proses, visualisasi matriks normalisasi.	Authenticated
Hasil & Laporan (/results)	Table ranking akhir, chart visualisasi (Bar/Radar Chart), tombol Ekspor PDF / Excel.	Authenticated

4. Alur Kerja Sistem & Data Flow

A. Alur Autentikasi Pengguna

1. **User Sign Up/Login** -> Mengisi form di Next.js Frontend.
2. **Supabase Auth** memverifikasi kredensial dan mengembalikan **JWT Session Cookie**.
3. **Next.js Middleware** membaca cookie di server side:
 - o Jika tidak valid -> Redirect ke /login.
 - o Jika valid -> Diizinkan masuk ke rute /dashboard/*.

B. Alur Pengolahan & Kalkulasi Data (Server Action Pattern)

1. **Input Data:** Pengguna mengisi kriteria, bobot, dan alternatif pada tabel di /data.
2. **Submit via Server Action:** Frontend memanggil Next.js Server Action untuk menyimpan data batch ke Supabase (projects, criteria, alternatives, evaluations).
3. **Eksekusi Perhitungan:**
 - o Pengguna mengklik tombol "Hitung / Analisis".
 - o Backend Next.js mengambil data matriks evaluasi dari Supabase.
 - o Algoritma dijalankan di Server Node.js (aman, tidak terekspos di browser client).
 - o Hasil kalkulasi (skor & ranking) disimpan kembali ke tabel results.

4. **Output:** Frontend menampilkan hasil peringkat, grafik chart, serta memberikan opsi ekspor laporan (PDF).

5. Panduan Konfigurasi Vercel (Deployment)

1. **Environment Variables** yang wajib diisi di Vercel Settings:
 - `NEXT_PUBLIC_SUPABASE_URL`: (URL Supabase Project)
 - `NEXT_PUBLIC_SUPABASE_ANON_KEY`: (Public Anon Key)
 - `SUPABASE_SERVICE_ROLE_KEY`: (Key khusus backend server/admin)
2. **Build Settings:** Standard Next.js framework preset (`npm run build`).

RINGKASAN DATABASE:

company_profiles: Identitas toko DynoBoo (logo, nama, kontak, alamat) untuk cetakan invoice.

invoice_types: Master template invoice (Workshop, Produk Rajut, dll) beserta prefix penomoran.

invoice_counters: Penyimpan urutan nomor invoice otomatis per bulan.

item_types: Kategori item (Workshop vs Produk Fisik).

items: Master katalog produk rajut dan tiket workshop.

stocks: Pengelola sisa stok/kuota (`qty_available`) dan jumlah terjual (`qty_sold`).

invoices: Header transaksi invoice (nama pembeli, total tagihan, status warna).

invoice_items: Detail rincian daftar barang yang dibeli per invoice (Parent-Child).

payments: Log riwayat pembayaran (DP / Pelunasan).